



Steady-state and power transient critical heat flux experiments of FeCrAl and Zircaloy claddings under saturated pool boiling

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ABSTRACT

FeCrAl and Zircaloy cladding tubes are vertically placed in the saturated pool boiling chamber to perform steady-state and power-transient critical heat flux (CHF) experiments. Three FeCrAl-variants, including FeCrAl-C26M, FeCrAl-B126Y, and FeCrAl-B136Y, are tested to compare their pool boiling thermal performances with traditional Zircaloy claddings including Zirlo, Zr705, and Zircaloy-4. In the steady-state pool boiling experiments, it was confirmed that FeCrAl alloys have higher pool boiling CHF values than Zircaloys. Among these tested materials, FeCrAl-C26M has the highest pool boiling CHF, but the FeCrAl-C26M nucleate boiling heat transfer coefficient is less than that of FeCrAl-B136Y. It reconfirms that CHF is enhanced on the post-CHF oxidized samples. However, as the cycling number of post-CHF oxidation increases, the CHF enhancement becomes less appreciable due to the limited physics contributions of the oxide layer. It is noteworthy that the finalized CHF enhancement can vary a lot between different cladding materials. For example, in three Zircaloy variants, their saturated pool boiling CHF results converge to 980 kW/m² on the post-CHF oxidized samples. In three FeCrAl variants, their pool boiling CHF results are finalized to 1400 kW/m². This CHF enhancement difference could be possibly attributed to the chemical constituents of oxide layers presented on cladding surfaces. The Fuchs-reactivity initiated accident transients are simulated by direct current programmable power supply to characterize thermal responses of different claddings to various pulse-width power transients. It is found that the heat transfer coefficients show monotonically increasing parametric trend with respect to faster power-transient rates, i.e., shorter pulse widths. Different from the steady-state pool boiling, the transition boiling regime is phenomenologically reflected in the power-transient pool boiling curve. This potentially implies that power-transient CHF featured by surface temperature overshoot may indicate transient thermal-safety margins not so accurately as the power-transient maximum heat flux. The difference gap of power-transient maximum heat flux between various cladding materials can be gradually closed by the progressive increasing of power transient rates.

1. Introduction

The experimental and computational studies of accident tolerant fuel (ATF) concepts have been on ongoing progress to support their near-term deployment in light water reactors (LWRs). The primary objective of ATF studies is to support the safe replacement of the traditional zirconium-based claddings by ATF alternatives including Cr-coated Zircaloys, FeCrAl alloys, and SiC/SiC_f. Insofar, a lot of ATF claddings thermal-hydraulics (T-H) assessments have been made to advance their loading in LWR cores since their advent (He et al., 2021b). Besides ATF cladding advantages in material mechanics, structural integrity, corrosion resistance, and hydrogen releasing mitigation, their T-H superiority over zircaloys have been also proven in pool/flow boiling experiments including critical heat flux (CHF), heat transfer

coefficients (HTCs), and minimum film boiling temperature (T_{MFB}) (He et al., 2021b; Seong et al., 2022; Moreira et al., 2022; He and Chen, 2022). Now, it is widely accepted that claddings' thermal-physical properties and wall thickness can exert pronounced impacts on their T-H performances. These impacts are most dominant under the saturated pool boiling (He et al., 2022; Yeom et al., 2020) while the progressive increasing of mass flux and/or liquid subcooling can reduce T-H performance dependency of ATF claddings on their material-related factors including thermal-physical properties/wall thickness, and surface morphological features (roughness, coating and oxide layer) (Moreira et al., 2022; He and Chen, 2022; Lee et al., 2020a, 2021).

Most of recent pool boiling experimental studies are performed for ATF and traditional claddings using plain plates. Ali et al. (2018, 2020) presented saturated pool boiling CHF experimental data points of

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Nomenclatures and Acronyms

q''	Heat Flux (kW/m ²)
V	Test Section Voltage (Volt)
$q(t)$	Test Section Thermal Power (Watt)
q_m	Peak Thermal Power (Watt)
s	Power Transient Time Scale (Second)
t	Time of Interest (Second)
T	Temperature (K)
C_p	Isobaric Specific Heat Capacity (J/(kg K))
k	Thermal Conductivity (W/(m K))
ρ	Mass Density (kg/m ³)
h	Specific Enthalpy (J/kg)
ϕ	Surface Orientation Angle (°)
θ_r	Receding Contact Angle (°)
ρ	Mass Density (kg/m ³)
ΔH_{fg}	Latent Heat of Evaporation (J/kg)
σ	Liquid Surface Tension (N/m)
G	Gravitational Acceleration Constant (m ² /s)
S	Non-dimensional Geometrical Factor (–)
L	Heated Length of Rod (m)
D	Outer Diameter of Heater Rod (m)
D_c	Capillary Length of Liquid (m)
ATF	Accident Tolerant Fuel
BWR	Boiling Water Reactor
CHF	Critical Heat Flux (kW/m ²)
DC	Direct Current
HL	Effective Heated Length (m)
HTC	Heat Transfer Coefficient (kW/m ² K)
ID	Inner Diameter (m)
LWR	Light Water Reactor
OD	Outer Diameter (m)
ONB	Onset of Nucleate Boiling (kW/m ²)
NB	Nucleate Boiling
PWR	Pressurized Water Reactor
RIA	Reactivity-Initiated Accident
SP	Single Phase
T-H	Thermal-Hydraulics

FeCrAl-C26M, bare and Cr-coated Zircaloy-4 that are irradiated and oxidized in the LWR simulated environments. Their experimental results demonstrated that FeCrAl-C26M gives better thermal performances including CHF and HTCs than Zircaloys, and irradiated and oxidized samples render higher CHF values than their non-pre-oxidized/irradiated counterparts. Most other pool boiling studies of ATF cladding plates are summarized by He et al. (2021b) outline how various factors of cladding plates affect pool boiling HTCs and CHF. However, it is impossible to directly extrapolate these pool experimental results to thermal safety assessments of LWRs because of boiling scheme difference.

Up to now, many ATF claddings boiling heat transfer experiments have been centered at the steady-state heat input scenarios. However, the power transient experimental studies of ATF cladding boiling heat transfer (Lee et al., 2019; He et al., 2023) are still too limited to characterize their T-H responses to scenarios progression of reactivity initiated accidents (RIA). Lee et al. (2019), Lee and Brown (2021) demonstrated that under the power-transient heat inputs, the traditional CHF concept that is featured by the cladding surface temperature overshoot is no longer applicable to thermal safety margin evaluation and the slug/churn flow conditions can result in the premature CHF occurrence of power transient. These experimental observations were

later confirmed by He et al. (2023). Moreover, He et al. (2023) experimentally solidified that the increasing of mass flux/inlet subcooling can suppress the effects of cladding material properties, wall thickness, and transient time scales on thermal-responses of cladding to the power-transient heat inputs. Although there have been accumulated a lot of power-transient boiling experimental results using various cladding materials and different heat input modes (He et al., 2021a), most of power-transient boiling experiments were conducted based on the exponential escalating heat input modes. This possibly leads to extrapolation difficulty even failure of those accumulated experimental results to full-scale LWR fuel rod bundle assessments. Because transient timescales together cladding thermal-physical properties have been shown significantly mechanistic effects on T-H responses of claddings to transient modes.

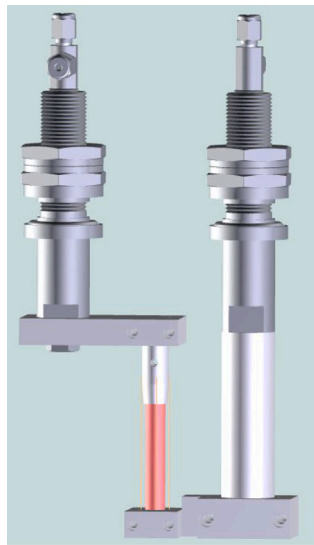
To complement T-H experimental evaluations of ATF claddings for better support their core-loading, the steady-state and power-transient saturated pool boiling experiments are performed upon three FeCrAl alloys and three Zircaloys respectively in this study. The rest of this paper is organized as follows: Section 2 elaborates experimental methodologies on addressing pool boiling facilities, tested samples preparations and metrological strategies. Section 3 presents the experimental results under the steady-state and power-transient heat input configurations in details. Results discussions are also made in Section 3 to articulate the experimental observations and facilitate the understanding of material-conjugate boiling heat transfer. And concluding remarks are made to summarize key points of this experimental study and the relevant implications are presented to the comparative assessments of ATF and traditional claddings in Section 4.

2. Experimental methodologies

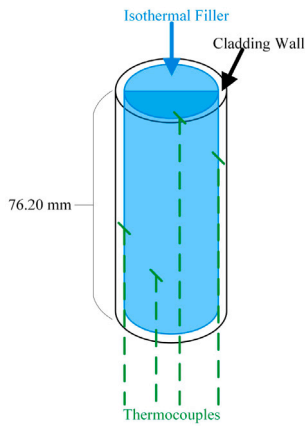
2.1. Pool boiling chamber and test section

Both steady-state and power transient experiments are conducted in the saturated pool boiling chamber. More details about boiling chamber can be found in authors' previous experimental studies (He et al., 2022). The only difference is to replace the horizontally-placed test section by the vertically-placed one (See Fig. 1(a)). Cladding tubes are clamped by customized copper parts and release the heat flux via outer surface based on the direct joule heating. The overall length of cladding tube is 76.20 mm while the effective heated length is 50.80 mm. The 12.70-mm-long residuals are tightly fastened on the cylindrical grooves of power feedthroughs using the copper clamps. A voltage sensor is connected to two terminals of copper clamps to measure the voltage drop across the cladding tubes while two current transducers are mounted upon power transmission cables to measure the DC power supply current. According to the user manual of OMEGA Engineering Inc, the uncertainties of voltage and current transducers are 2.65% and 8.47% respectively at their maximum allowable scales.

Three FeCrAl variants and three Zircaloys are used to characterize their thermal responses to two different heat input conditions. Their outer diameter, wall thickness, surface roughness and contact angles can be found in He et al. (2022). All the tested samples involved in this experimental study are prepared following exactly same procedures to He et al. (2022) only except thermocouple placements. As shown in Fig. 1(b), the effective heated region is partitioned in 4 length-identical regions and 4 temperature thermocouples (Item# SA1-K from Omega Engineering Inc.) are placed upon the inner surface of cladding tube at the region center. And four thermocouples are distributed radially at 12, 3, 6 and 9 O-Clock directions respectively. The SA1-K thermocouple has the minimum response time of 0.3 s to the 63.2% of an instantaneous temperature change. This can enable the SA1-K thermocouple to capture the temperature escalating rate at the maximum of 368 K/sec. Thus, the thermal inertia impacts on power transient results are negligible in our study. Given that the steady-state and power transient boiling experiments are conducted under the saturated pool



(a) Position



(b) Configuration

Fig. 1. Vertically-placed test section posing cladding tubes that are marked by light red.

boiling condition. Basically, the pool boiling chamber is at the saturated temperature of deionized water, and the system configuration and layout do not block the adjacent water supply to re-wet outer surfaces of the vertically-placed cladding tubes. On the contrary, the copper clamps that transfer electricity passing through the test section would slightly pose extra thermal impacts on the metrological results. However, the total electrical resistance of test section is much larger than that of copper clamps. The direct joule thermal power of copper clamps is infinitesimal in comparison with the total thermal power of test section. Therefore, the influences of experimental apparatus on inner surface temperature measurements and bubble behaviors can be neglected. Noting that the hollow space of cladding tube is filled by liquid cement that has relatively higher thermal expansion coefficient in comparison with other concrete cement products, and both ends are sealed by high-temperature tolerant epoxy. To ensure robust thermal bonding of four thermocouples against the cladding tube wall, this study adopts the following techniques, (1) strongly adhesive thin pads (0.012 mm-thick) are used to stick thermocouples upon the inner surface tightly, liquid cement is filled in the hollow space of cladding tube 10 droplets ($10 \times 50 \mu\text{L}$) per each operation to form a cement layer, then a SS 316 rod (Diameter 3 mm) is used to squeeze air voids out of the cement layer, and the cladding tube samples are placed in the dry and room temperature hood at least for 2 days for the cement dehydration and

solidification, and then repeat this procedure until the hollow space is fully stuffed with cement base. Also the cement filler expansion under high temperature can help push thermocouple terminals better against wall to measure the inner surface temperature. Thermal inertia effects of $3 \times 3 \times 0.012 \text{ mm}^3$ adhesive pads on measurements could be negligible because of the thin wall.

The deionized water is degassed by such strategy that the saturation temperature of temperature is kept by two wall heaters working at least until there are no bubbles generated on the test section or chamber walls. Such this status is then kept for 45 min before subjecting test samples to heat inputs.

2.2. Heat input strategies

In this study, both steady-state and power-transient heat input modes are implemented by DC programmable power supply (Magna-Power Electronics TSA-16-1200) to probe saturated pool boiling behaviors of FeCrAls and of Zircalloys respectively. In order to mitigate the potential damage of joule heating open circuit on the power supply unit, this study adopts the voltage-based control mode to produce both the steady-state and power-transient heat inputs. The step-by-step incremental strategy of voltage is used herein to generate the quasi-state thermal power as an alternative approximation of the steady-state heat input, as follows,

$$V(k+1) = V(k) + \Delta V \quad (1)$$

where $V(k)$ is the voltage drop across the tested samples at the k th steady-state stage, ΔV is the voltage incremental step (~ 0.01 volt in this study). The steady-state signifier of test section is that two time-wise mean temperature points have no difference beyond 1°C provided that a time-wise mean temperature point is averaged over 50 consecutive temperature points from the same signal channel. Before increasing the power voltage to the next level, the previous steady-state stage is kept at least for 40 s

The raised cosine distribution is employed to approximate the Fuchs-RIA power-transient mode as follows

$$q(t) = \begin{cases} \frac{q_m}{2s} [\cos(\pi \frac{t-s}{s}) + 1] & t \in [0, 2s] \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

where s is the power-transient timescale, t is the time domain, and q_m is the peak thermal power of cladding rod. In Eq. (2), s characterizes the changing rate of heat inputs. The primary rationale behind Eq. (2) approximation to Fuchs-RIA is that it is more convenient to generate faster signals that are fed to power supply without truncating the $\text{sech}^2(t)$ tails or delaying the slew rate response of power rotary circuit. q_m is fine tuned by the peak voltage of control signal. The waiting time between two consecutive occurrences of power transients is 2 min. This allows the tested section to be completely cooled down and to return to steady state prior to the next subsequent transient. The absolute difference between two time-wise average temperature points is beyond 30°C or any temperature signal channel reads the inner surface temperature beyond 400°C . This implies that CHF occurs on the cladding surface, or the boiling regime transits to the film boiling. Once either CHF or film boiling is detected, the power supply will be automatically shut down.

3. Experimental results and analyses

3.1. Steady-state cases

New experimental results of ATF and traditional claddings are presented herein under the vertically-placed saturated pool boiling scheme and to complement the enlarging database of ATF T-H experimental evaluation (He et al., 2022, 2021b). Given that thermocouples are placed upon the inner cladding surface to procure temperature responses and the wall thickness impacts on heat conduction are not

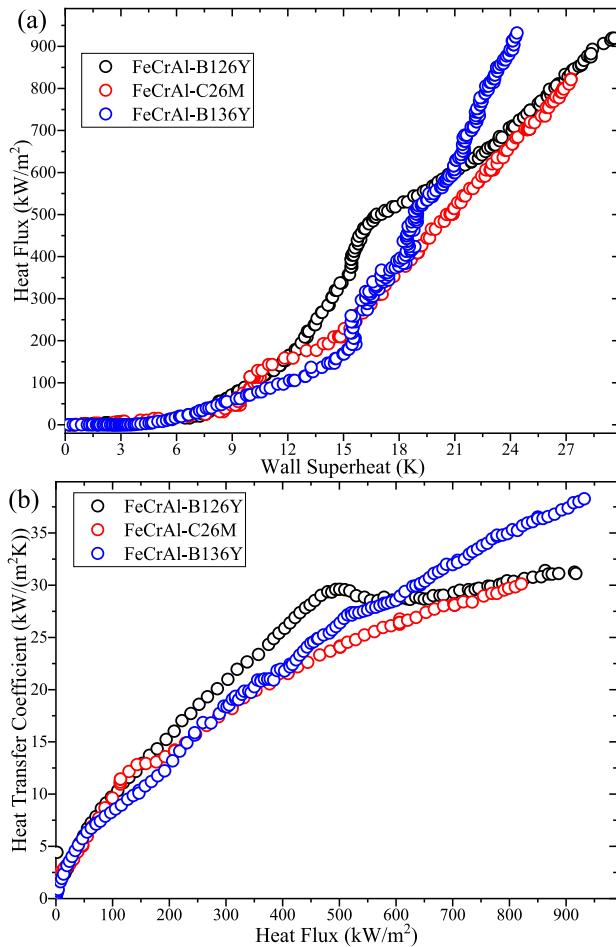


Fig. 2. Experimental results of three FeCrAl alloys (a) pool boiling curves and (b) heat transfer coefficients, C26M, B126Y and B136Y.

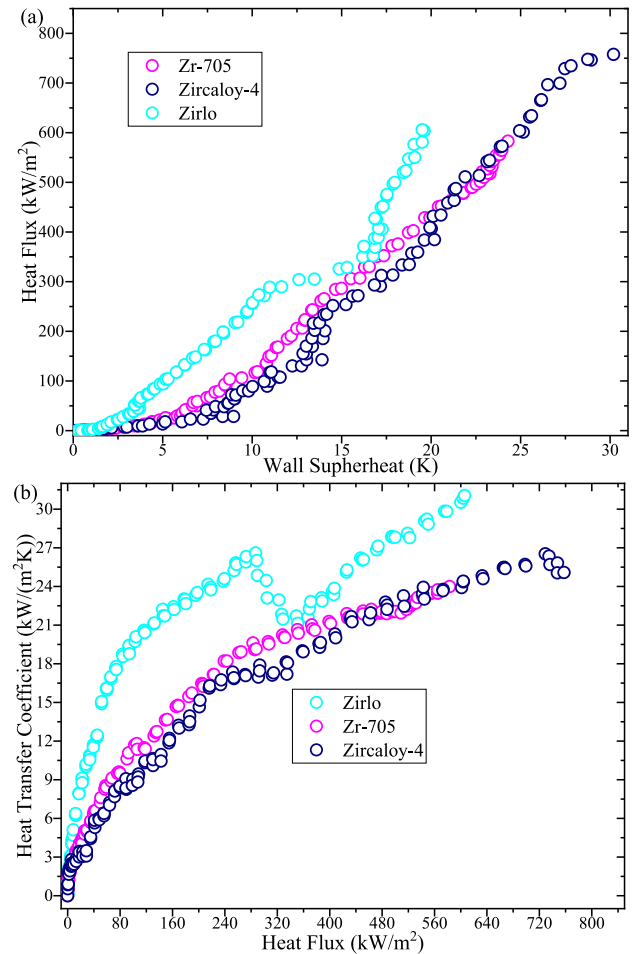


Fig. 3. Experimental results of three Zircaloy alloys (a) pool boiling curves and (b) heat transfer coefficients, Zircaloy-4, Zirlo and Zr-705.

negligible, the outer surface temperature is calculated via the inner surface temperature measurements via the steady-state 1D heat conduction model (See Appendix A in He et al. (2022)). Fig. 2 and Fig. 3 respectively present the pool boiling experimental results of three FeCrAl alloys and three Zircaloy alloys. The vapor quality increases axially upward along the rod height. Surface temperature overshooting is more likely to occur at the topmost region of tested sample. The inner surface temperature procured by the thermocouple being placed at the topmost region is the input parameter of interest to the steady-state 1D heat conduction model. Therefore, the pool boiling heat transfer coefficients are evaluated at the topmost region.

In terms of natural circulation heat transfer coefficient (NB-HTC), onset of nucleate boiling (ONB), and nucleate boiling heat transfer coefficient (NB-HTC) between three FeCrAl variants, experimental observations under the vertically-placed schemes are much similar to the authors' previous studies of horizontally-placed ones (He et al., 2022). Still, NB-HTC of FeCrAl-B136Y is higher than that of FeCrAl-C26M and FeCrAl-B126Y. This agrees with the authors' previous studies (He et al., 2022). Only difference is that HTC and ONBs procured under the vertically-placed schemes are higher than their counterparts under horizontally-placed ones. The mechanistic rationales behind such this difference are bubbles under the vertically-placed scheme are more likely to detach from the cladding than under the horizontally-placed scheme. It was pointed out by Chuang et al. (2019), Goel et al. (2017) in their experimental studies that the bubble departure frequency and diameter increase as surface orientation (ϕ) varies from 0° to 90° (upward-facing surface) because of bubble induced turbulent disturbance. On

the other hand when ϕ increases from 90° to 180° (downward-facing surface), the tendency of heating surface prohibiting bubbles crowds detachment from itself become more and more obvious. This implies that 90° maximizes the bubble departure frequency. In our pool boiling experiments, the ϕ of the horizontally-placed tubes varies from 0° to 180° continuously while the ϕ of the vertically-placed tube is always fixed at 90° . Therefore, it is reasonable to postulate the averaged bubble departure frequency on horizontally-placed tube less than that of vertically-placed tubes. This may physically attribute to higher NB-HTC under vertically-placed pool boiling schemes. As shown in Fig. 3, experimental observations between three Zircaloy variants in this study are also much similar to authors' previous study using horizontally-placed tube. More related discussions can be found in He et al. (2022).

Fig. 4 presents pool boiling experimental CHF results of six cladding materials of interest under both horizontally- and vertically-placed schemes. As observed in Fig. 4, the experimental CHF results using vertically-placed tubes are higher than that of horizontally-placed tubes. Compared with the vertically-placed schemes, the horizontally-placed tubes are more susceptible to bubble coalescence irreversibly into a vapor layer. As observed in the horizontally-placed pool boiling experiments (He et al., 2022), CHF firstly occurs at the downward face, and then expands to the upward face. This is due to bubbles from the upper half and lower half of tube tend to merge and form vapor columns (Rao and Andrews, 1976). However, for the vertically-placed schemes, its bubble departure frequency is higher than the average one of horizontally-placed tubes. The micro-layer CHF model and improved

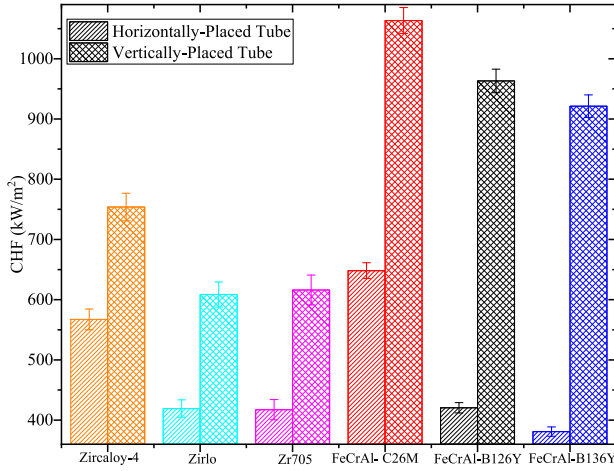


Fig. 4. Pool boiling CHF experimental results of three as-received FeCrAls and three as-received Zircaloys under horizontally/vertically-placed schemes.

variants (Liu et al., 2019; Kim et al., 2015) point out that pool boiling CHF is proportional to bubble departure frequency. Still FeCrAl-C26M outperforms other 5 cladding materials in the vertically-placed pool boiling.

In the actual LWRs, cladding materials are exposed to high-temperature high-pressure coolants and are bombarded by radioactive particles, resulting in the formation of oxide layer on outer cladding surface. The impacts of oxide layer on pool boiling CHF have been proven appreciable in many experimental studies of oxidized ATF clads (He et al., 2021b). In this study, the following strategy is implemented to study the impacts of oxide layer on pool boiling CHF of ATF and traditional claddings in an accelerated manner. Considering that surface temperature overshooting can abruptly oxidize the boiling heat transfer surface, the steady-state pool boiling experiments are repeated on the same test specimens (post-CHF surfaces) until the specimens are damaged or the thermocouples cannot procure inner surface temperature correctly. Such these procured experimental results can partially reflect the impacts of oxide layer on pool boiling CHF. As shown in Fig. 5, the pool boiling experimental CHF of three FeCrAls and three Zircaloys variants increases as over the progressive increasing of cycle number. However, the further increasing of cycle number cannot enhance pool boiling CHF anymore. This speaks to the possibility that the oxide layer enhancement on pool boiling CHF is limited. The plausible rationales behind this observation are (1) surface morphology, chemical constituents, thickness of oxide layer become stabilized, and (2) the further increasing of oxide layer thickness has asymptotic enhancement on pool boiling CHF (similar to the impacts of wall thickness on pool boiling CHF, See Guglielmini and Nannei, 1976; Arik and Bar-Cohen, 2003). It is noteworthy that the presence of oxide layer can diminish the difference gap of pool boiling CHF between FeCrAl (Zircalloy) variants. For example, at the cycle number of 2, experimental CHF data points procured on post-CHF samples of three FeCrAls (or Zircaloys) show no appreciable difference in comparison with the notable gap between as-received samples. Given that there is no significant difference among FeCrAls and Zircaloys in terms of surface roughness (Find Table 4 in He et al., 2022), thermal-physical properties of claddings together with surface wettability could account for the CHF difference gap between as-received “Zircaloys” samples. If the cladding surface is fully oxidized, surfaces become more hydrophilic and oxide presences are closely same. On the other hand, three fully oxidized FeCrAl variants have a higher pool boiling CHF value than three oxidized Zircaloys. From parametric trend of cycle number with respect to pool boiling CHF in Fig. 5, the fully oxidized samples from the same alloy family may have same pool boiling CHF

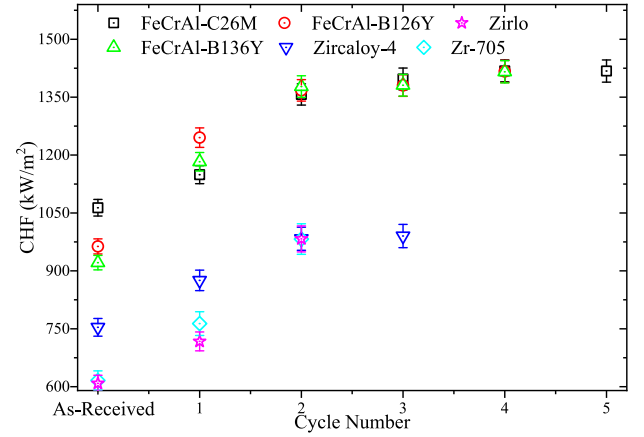


Fig. 5. Pool boiling CHF experimental results of ATF and traditional claddings between as-received and post-CHF surface.

as they share similar chemical element compositions and contents. Otherwise, presence of different oxide constituents may induce significant CHF difference gap for instance, in Fig. 5, the finalized CHF of FeCrAls is ~430 kW/m² higher than that of Zircaloys.

3.2. Power-transient cases

The 1D transient heat conduction equation with volumetric source term is used to predict the outer surface temperature based on the inner surface temperature measurements as follows

$$\rho C_p(T)_w \frac{\partial T}{\partial t} = \frac{\partial}{\partial r} [k(T)_w r \frac{\partial T}{\partial r}] + \frac{4q(t)}{\pi(OD^2 - ID^2)HL} \quad (3)$$

where $C_p(T)_w$ and $k(T)_w$ are cladding specific heat capacity and thermal conductivity that are weakly dependent of temperature, OD and ID are the outer and inner diameters of cladding tube, HL is the effective heated length of cladding (HL is 50.80 mm in this study), and $q(t)$ is the thermal power of heat input that is measured by voltage and current transducers. For the cement filler, the transient heat conduction model is as follows,

$$\rho C_p(T)_s \frac{\partial T}{\partial t} = \frac{\partial}{\partial r} [k(T)_s r \frac{\partial T}{\partial r}] \quad (4)$$

where $C_p(T)_s$ and $k(T)_s$ are cement specific heat capacity and thermal conductivity. The boundary conditions of Eqs. (3) and (4) are written as follows,

$$T(r = \frac{ID}{2}, t) = T_c(t) \quad (5a)$$

$$-k(T)_s \frac{\partial T}{\partial r} \bigg|_{r=0} = 0 \quad (5b)$$

where $T_c(t)$ is the measured inner surface temperature at the topmost region of tested sample. The Crank–Nicolson finite difference method is adopted to numerically solve Eqs. (3) and (4) and construct power transient pool boiling curves due to its unconditional stability.

With the assist of the computational scheme of Eq. (3), Eq. (4) and (5), the power transient pool boiling curve can be constructed based on the inner surface temperature and thermal power input measurements. Fig. 6 illustrates the inner surface temperature measurement to a complete Fuchs-RIA power pulse acting on the FeCrAl-B126Y cladding. As shown in Fig. 6, the maximum inner surface temperature of cladding material is delayed in appearing in the reflected thermal response profile to the Fuchs-RIA power transient when compared with the peak heat flux of the cladding material. This can be mechanistically explained by the transient energy deposition in the cladding material wall.

The power-transient boiling curve could be constructed to reflect the thermal response of FeCrAl-B126Y to a complete Fuchs-RIA power

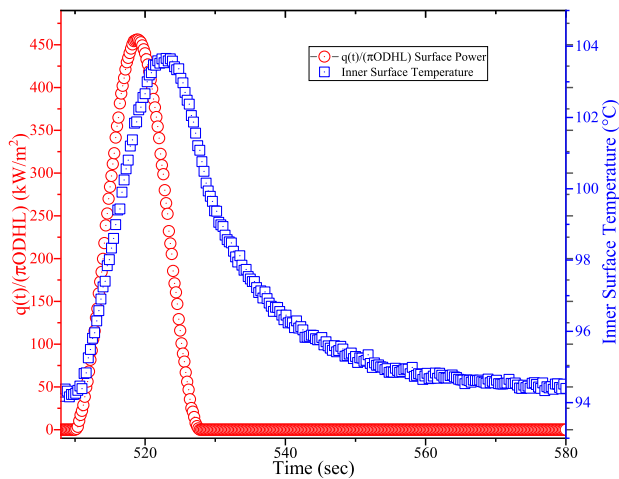


Fig. 6. FeCrAl-B126Y inner surface temperature response to a complete “ $s = 8$ s” Fuchs-RIA power pulse.

pulse by feeding measured data of Fig. 6 to the computational framework. The corresponding output data sets are reflected by the outer surface temperature and heat flux (See Fig. 7(a)). Due to the deposited thermal energy releasing back to the coolant, the outer surface temperature responses to the increasing stage is phenomenologically different from the decreasing stage as reflected in Fig. 7(a). This further results in the increasing stage HTC much higher than that of the decreasing stage (See Fig. 7(b)). Besides, the single phase natural circulation regime is reflected during the power-increasing stage while not during the power-decreasing stage. It is also confirmed that power transient HTC are higher than steady-state HTCs. Fig. 8 shows that bubble dynamics behaviors on the vertical placed FeCrAl-B126Y tube at seven heat flux points whose time indexes are symmetrically distributed. Because of the thermal deposited energy releasing to coolant, the heat flux of power-decreasing stage is slightly higher than the heat flux of power-increasing stage at two timing-symmetric points. It could be found in Fig. 8 that the bubble crowd density during the power-decreasing stage is higher than that of the power-increasing stage. Given that HTCs are evaluated at the topmost region of uniformly heated tube, local bubbles together with the elevated bubbles that are detached before could result in the previous bubble dynamics having suppressing impacts on the latter-following bubble behaviors. This may partially rationalize the higher HTC of power-increasing stage.

Fig. 9 presents the experimental HTC results of FeCrAl-B126Y under the complete Fuchs-RIA power pulses with different transient time scales. As shown in Fig. 9, NB-HTCs of four different transient time scales are ranked as $s = 2$ s > $s = 6$ s > $s = 8$ s > $s = 4$ s while for NC-HTCs, it is sequence as $s = 6$ s > $s = 8$ s > $s = 4$ s > $s = 2$ s. These observations are in confirmation with the experimental results of power-transient pool boiling (He et al., 2021a) that higher power-transient slew rates could lead to lower NB-HTC. This speaks to that heat transfer mechanisms across the triple surface are contributed by at least heat conduction/convection. Regarding to how heat conduction/convection act upon the mechanistic mechanisms, it still remains vague and unsolved as there are a few power transient pool boiling experimental studies investigating bubble dynamics (Scheiff et al., 2019, 2021).

Given that FeCrAl-C26M and Zircaloy-4 are representative ATF and traditional cladding materials, their power transient pool boiling HTCs are evaluated at the topmost region of heated tubes. Fig. 10 presents pool boiling experimental HTC results of FeCrAl-C26M and Zircaloy-4. Also, the pool boiling hysteresis is obviously more evident when heat flux approaches to the onset of nucleate boiling (ONB) under the

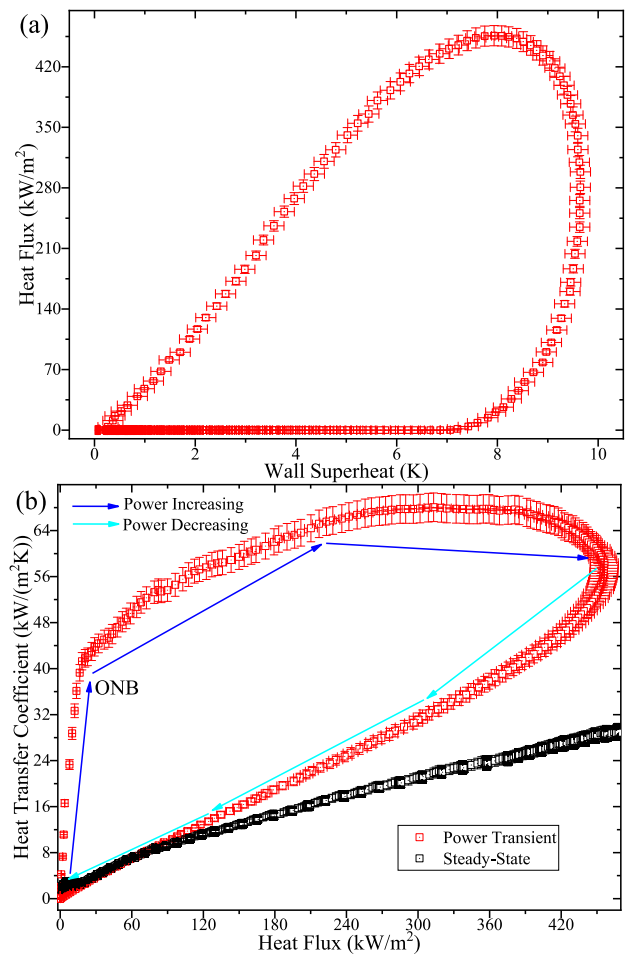


Fig. 7. Thermal responses of FeCrAl-B126Y to Fuchs-RIA power pulse $s = 8$ s: (a) pool boiling curve (b) heat transfer coefficients.

subjection to Fuchs-RIA power pulses due to NC-HTC higher than NB-HTC around the ONB. It is observed in Fig. 10(d) for the $s = 8$ s complete Fuchs-RIA pulse, the Zircaloy-4 NB-HTC during the power-decreasing stage shows same parametric trends to that of FeCrAl-C26M with respect to heat flux. And also the Zircaloy-4 NB-HTC during the power-decreasing stage behaves similarly with that of FeCrAl-C26M in the $s = 2$ s Fuchs-RIA case (See Fig. 10(a)). However, under the $s = 4$ s and $s = 6$ s power transient cases (See Figs. 10(b) and 10(c)), under the power-decreasing stage, the NB-HTC difference gap between Zircaloy-4 and FeCrAl-C26M converges as the power decreasing from peak to the zero. As pointed out in He et al. (2021a), under very slow power transients, T-H responses of cladding materials show no appreciable difference from the steady-state heat input conditions while under the fast power transient, T-H responses of claddings could be approximated by transient heat conduction. But under the immediate power transient rate, there are some unknown heat transfer mechanisms dominated behind CHF and HTC. This is in a good agreement with the experimental observations in Fig. 10.

Fig. 11 presents surface temperature overshooting phenomena reflected on FeCrAl-B136Y under the subjection to a $s = 8$ s Fuchs-RIA power pulse. Due to the automatic detection of surface temperature overshooting, the power supply system is immediately shutdown. One of obvious differences between the power transient and steady-state pool boiling is that transition boiling appears in the pool boiling curve, as can be seen in Fig. 11. Besides in pool boiling, such this boiling phenomenon is also reported in experimental studies of power-transient flow boiling (He et al., 2023). Considering this apparent

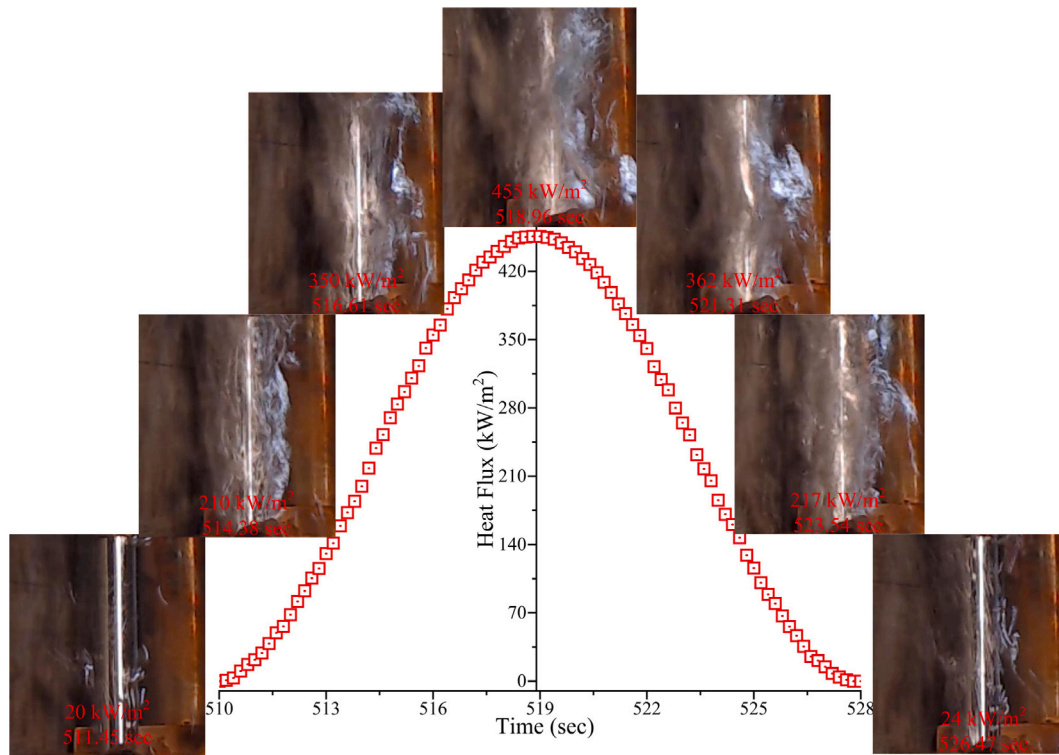


Fig. 8. Bubble behaviors of FeCrAl-B126Y at seven power transient heat flux points of a complete “ $s = 8$ s” Fuchs-RIA power pulse.

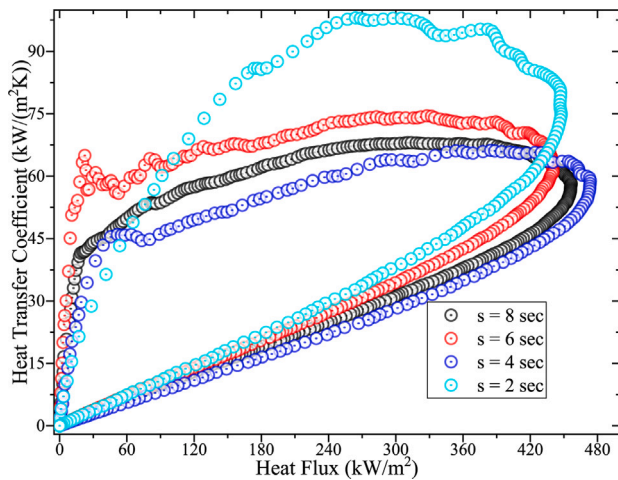


Fig. 9. Experimental HTC results of FeCrAl-B126Y under four different power transient time scales.

distinction between power transient boiling and steady-state boiling, the maximum heat flux (MHF) concept from He et al. (2023), is hereby adopted to quantify LWRs thermal safety margins of pool boiling under the RIA impacts. This implies that if the peak heat flux of a Fuchs-RIA power transient pulse is at least beyond MHF, surface temperature overshooting to film boiling will definitely occur on the cladding surface. Otherwise, the Fuchs-RIA power transient pulse will not induce surface temperature overshooting and will not pose thermal threats to the mechanical integrity of the fuel-clad entity. And more importantly, under the Fuchs-RIA power transient scenarios, MHF is a better thermal safety indicator of LWRs than CHF that is usually featured with surface temperature overshooting. Fig. 12 presents the experimental MHF results of three FeCrAls and three Zircalloys under four different power transient time scales, and compare them with

their steady-state CHF counterparts. The steady-state CHF values of six cladding materials are lower than their power-transient MHFs, which gives a solid confirmation to the concluding remarks made in He et al. (2021a). Besides, it is also experimentally validated in this power-transient pool boiling study that faster power transients lead to higher MHF. According to the literature review study of power transient (He et al., 2021a), the very slow power transient heat input conditions make bubble dynamics and T-H responses no distinct from the steady-state cases because the very slow power transient could be approximated by quasi-state or steady-state heat inputs. This speaks to the possibility that the use of present steady-state CHF tables can be conservatively applicable to the thermal safety margins of Fuchs-RIA accident progression scenarios in LWRs. The new experimental findings of this study are that (1) the progressive decreasing of transient time scale could gradually close the difference gap of power transient MHF between different claddings due to the gradual dominance of heat conduction over heat convection and (2) FeCrAl alloys can also provide higher thermal safety margins than Zircalloys under power transient heat input conditions because of the enhanced thermal conductivity of FeCrAls. It is reasonable that under the ultra-fast power transient cases, MHF values would be independent of cladding materials, and only dependent of thermal hydraulics instabilities and power-transient time-scales. Insofar, the only formally identified mechanism behind power transient CHF occurrence proposed by Sakurai (2000) was heterogeneous spontaneous nucleation that assumed the direct transition from single-phase heat transfer to film boiling without nucleate boiling. This assumption is only limited to extremely short-periods because surface power needs surging ultra-fast to turn adjacent liquids into vapor layer before the nucleating process of first bubble generations. Regarding to the experimental transient time scales of this study from 2 s to 8 s, heterogeneous spontaneous nucleation could partially rationalize why faster power transients lead to higher MHF. Because the surface power surges so fast that the adjacent liquid could be insufficient of contact time to be heated up before turning in vapor layers, and more thermal energy is needed to compensate the shorter contact time. Also on the other hand, the establishment of irreversible dry patch formation

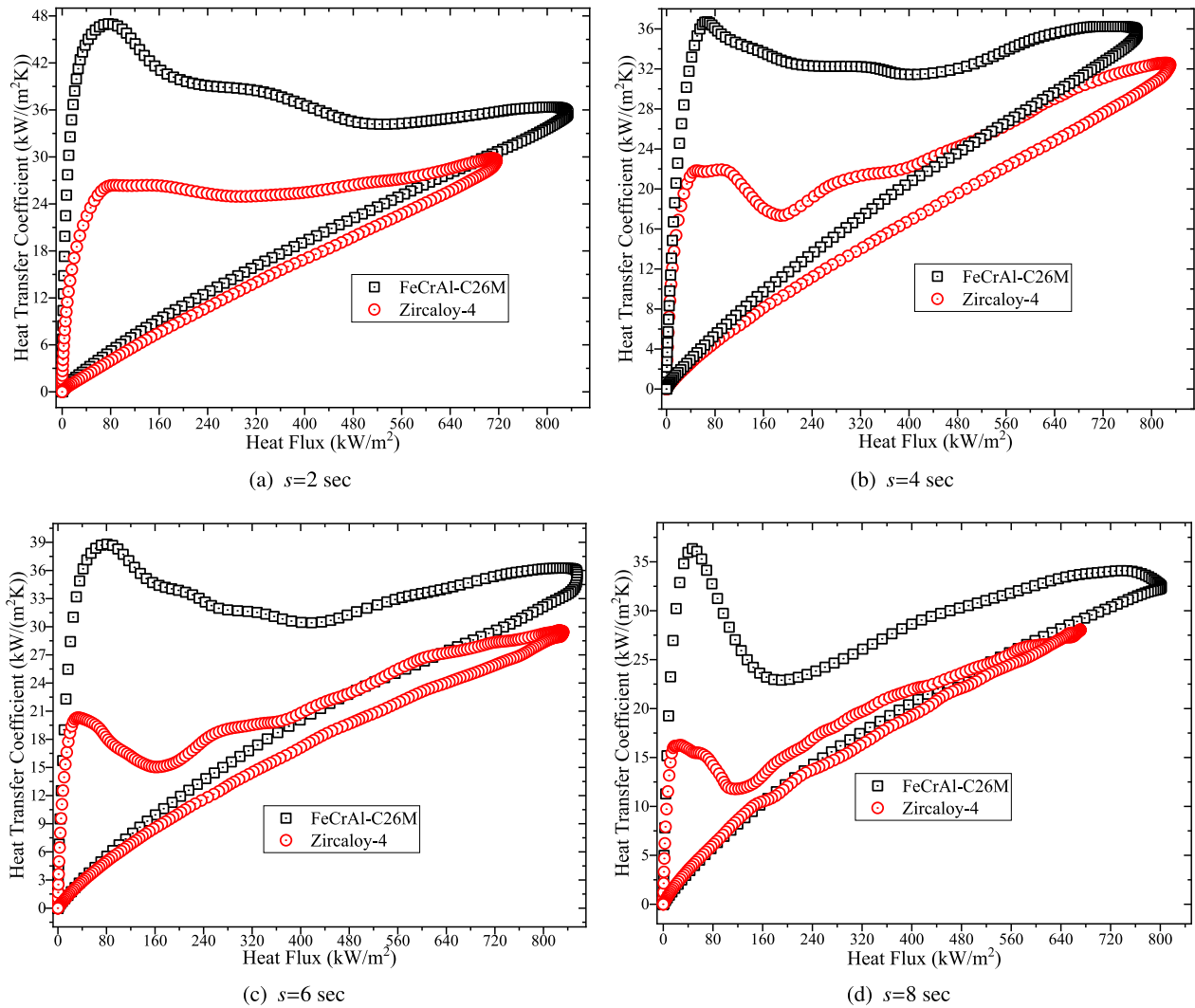


Fig. 10. Power transient pool boiling experimental HTC results of FeCrAl-C26M and Zircaloy-4 under four different complete Fuchs-RIA power pulses.

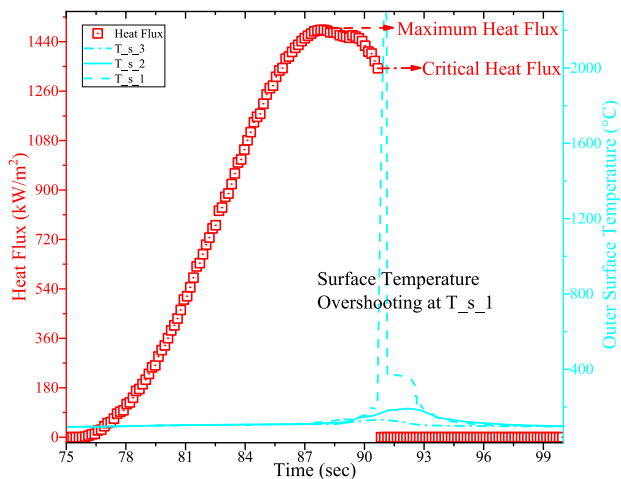


Fig. 11. Power transient pool boiling of FeCrAl-B136Y with surface temperature overshooting under the $s = 8$ s power pulse.

is related with the hydrodynamic instability based bubble dynamics. Right before the bubbles detaching from the surface, if the power surges up too fast to enable the adjacent liquid supply re-wetting the

local hot spots, the nearest nucleation spots could not be activated enough in time to form noticeable dry patches. Thus, higher thermal energy is demanded to activate suppressed nucleation sites. This may mechanistically faster power transients resulting in higher pool boiling MHF. Similar to steady-state CHF mechanisms, physics mechanisms behind power transient MHF/CHF still remain vague, incomplete and even controversial. This needs more advanced diagnostics techniques to probe power transient boiling heat transfer including high-speed infrared camera, stereo camera, and high-temperature silica fiber.

3.3. Results modeling and discussions

Monde et al. (1994), Monde and Yamaji (1990) proposed an semi-analytical pool boiling CHF model for the vertically-placed tubes that are submerged in saturated liquids based on their experimental data as follows,

$$q''_{CHF} = S \Delta H_{fg} \rho_g^{0.5} [\sigma G (\rho_l - \rho_g)]^{0.25} \quad (6)$$

where ΔH_{fg} is the latent heat of evaporation, ρ_g and ρ_l are vapor and liquid densities of working fluid respectively, σ is the surface tension of working fluid, and G is the gravitational acceleration constant. In Eq. (6), S is the geometrical factor that correlates impacts of

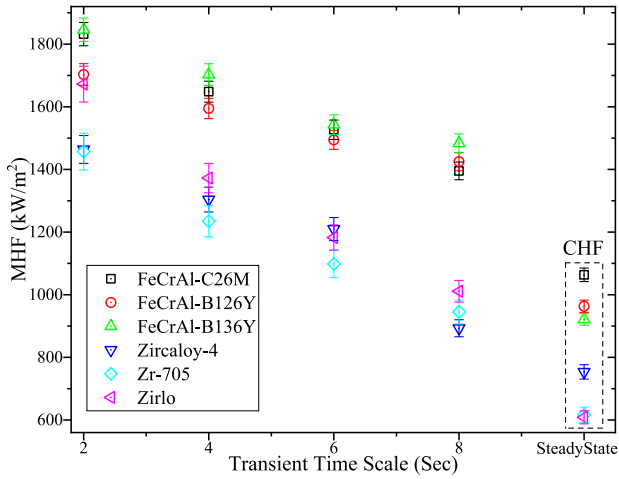


Fig. 12. Pool boiling MHF experimental results of three as-received FeCrAls and three as-received Zircalloys being subjected to different Fuchs-RIA transient scales.

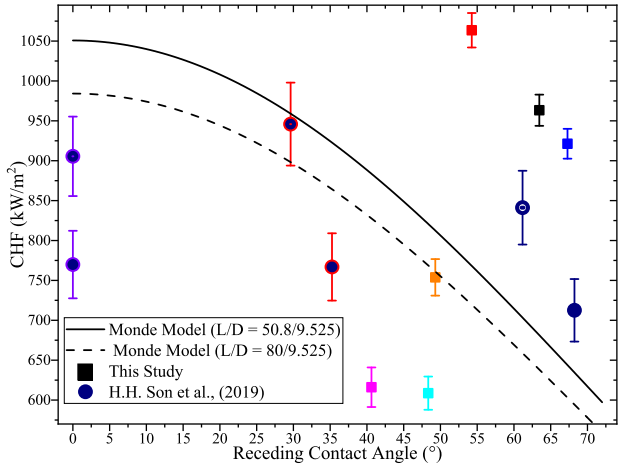


Fig. 13. Steady-state pool boiling CHF experimental results assessed by the improved Monde model of Eq. (8) noting that in the study of Son et al. (2019), the violet edge color denotes Cr-coated SS 316 while the red edge color denotes FeCrAl-coated SS 316, the dot-center circles represent tested samples having higher surface roughness.

vertically-placed tubes on pool boiling CHF, as follows,

$$S = \begin{cases} \frac{0.16}{1+0.025 \frac{L}{D}} \frac{D}{D_c} < 13 \\ \frac{0.16}{1+0.003 \frac{L}{D_c}} \frac{D}{D_c} \geq 13 \end{cases} \quad (7)$$

where L is the effective heated length of tube, D is the outer diameter of tube, and D_c is the capillary length of working fluid $\sqrt{\sigma/(G(\rho_l - \rho_g))}$. It should be noted that Eq. (7) is only applicable to the external pool boiling without confinement. To accommodate effects of surface contact angle on pool boiling CHF, Eq. (7) is modified based on the Kandlikar CHF model (Kandlikar, 2001), as follows,

$$S = \begin{cases} \frac{C[1+\cos(\theta_r)]\sqrt{\frac{2}{\pi} + \frac{\pi}{4}[1+\cos(\theta_r)]}}{1+0.025 \frac{L}{D}} \frac{D}{D_c} < 13 \\ \frac{C[1+\cos(\theta_r)]\sqrt{\frac{2}{\pi} + \frac{\pi}{4}[1+\cos(\theta_r)]}}{1+0.003 \frac{L}{D_c}} \frac{D}{D_c} \geq 13 \end{cases} \quad (8)$$

where θ_r is the receding contact angle, and C is the fitting constant.

Prior to this experimental study, Son et al. (2019) coated Cr and FeCrAl-APMT on SS 316 tubes, and experimentally characterized their steady-state pool boiling CHF on coated and bare SS 316 tubes under the vertically-placed scheme. Based on the steady-state pool boiling

experimental CHF data points of this study and Son et al. (2019), C is found 0.014 by the nonlinear regression technique. The steady-state pool boiling experimental CHF results of this study and Son et al. (2019) are used to assess the improved Monde model as shown in Fig. 13. As reflected by the improved Monde model, the increasing of vertical-placed tube heated length leads to lower pool boiling CHF because the bubbles that elevate axially upward along the tube accumulate around the topmost region and make the irreversible formation of dry patches more likely occur at the topmost region. Although the improved Monde model can correlate the impacts of heated length, surface wettability, the discrepancy between predictive and experimental results is still too appreciable to be modeled. It is revealed in the recent pool boiling ATF experiments that cladding thermal-physical properties may play dominant roles in steady-state CHF mechanisms (He et al., 2021b). Surface roughness and cladding thermal-physical properties should be incorporated in the future variants of pool boiling models to improve CHF predictability to various applications of boiling heat transfer.

3.4. UNM thermal-hydraulics experiments of ATF claddings: A brief look

There has been a systematic thermal hydraulics experimental evaluation of ATF cladding materials since 2018 including steady-state and power transient pool/flow boiling experiments, cladding quenching and water/steam corrosion studies at UNM. These relevant ATF claddings experimental studies are still on-going to support their large scale core deployment in LWR cores until the midst of 2020s.

Pool boiling experiments. Ali et al. (2018) presented the pool boiling CHF experimental results of the LWR autoclave pre-oxidized square plates $10 \times 10 \times 0.5 \text{ m}^3$ of two FeCrAl variants and Zircaloy-4 against their as-machined counterparts. They attributed the oxidation rate and characteristics of the oxide layer (thickness and microstructure) to higher pool boiling CHF of FeCrAl samples. Later, Ali et al. (2020) showed that the Cr-coated/pre-irradiated Zircaloy-4 samples render higher pool boiling CHF than the bare Zircaloy-4 sample because of surface roughness and wettability enhancement. He et al. (2022) systematically investigated how heater materials and wall thickness affect pool boiling CHF and heat transfer coefficient by using the horizontally-placed cladding tubes including stainless steels, nickel alloys, Zircalloys and FeCrAl alloys. These previous pool boiling experimental studies sufficiently prove that the use of FeCrAl claddings can boost thermal safety margin of LWR cores. However, due to the cladding material configurations in their experimental investigations, the related pool boiling CHF results can not directly be translated to the vertically-placed ATF fuel assembly. This study provides the experimental pool boiling CHF results of three FeCrAl alloys and three Zircalloys respectively using the vertically-placed configuration.

Flow boiling experiments. He et al. (2019) reported that there exists the difference gap of steady-state flow boiling CHF between stainless steel 316 and Zirlo due to the variations of cladding thermal physical properties. However, the progressive increasing of mass flux can suppress the effects of cladding thermal physical properties on flow boiling CHF difference gap. This was then experimentally confirmed using FeCrAl alloys, nickel alloys, stainless steels and other engineering materials, such that the gradual increasing of mass flux (Lee et al., 2020b,a; He and Chen, 2022) can reduce the difference gap of flow boiling CHF between various claddings and the higher liquid subcooling (He and Chen, 2022) also results in smaller difference gap. This speaks to the possibility that the existing CHF lookup tables procured by stainless steels and/or Inconels could conservatively be employed to assess thermal safety margins of ATF-loaded LWR cores under the nominal operations. Besides steady-state flow boiling experiments of ATF claddings, Lee et al. (2019) showed that the power transient flow boiling CHF of FeCrAl-C36M under different heating rates is much higher than that of their steady-state cases, later Lee and Brown (2021) showed that the

increasing power transients could diminish the influence of flow boiling heat convection and the volumetric heat capacity of cladding would be more significant under the fast power transients by the variance based global sensitivity perturbation studies. Also, He et al. (2023) illustrated that under the Fuchs-RIA power transients, flow boiling phenomena are completely different from the steady-state situations in terms of surface temperature overshooting from nucleate boiling to film boiling. It is more important that the flow boiling instability induced by the power transient can enhance boiling crisis oscillation but can be suppressed by increasing of mass flux and/or of inlet subcooling. The power transient pool boiling experimental parts in this work agrees with the previous power transient flow boiling for that power transient MHF is a better thermal safety indicator than the power transient CHF. Moreover, the experimental results of power transient MHF on this pool boiling study are in a solid confirmation that heat conduction tends to dominate over heat convection as over the increasing of power transient rate. This is complemented to the previous power transient flow boiling study (He et al., 2023).

4. Concluding remarks and summaries

4.1. Key summaries

Throughout our pool boiling experimental studies, the following conclusions are summarized as follows:

- (1) the vertical placement is experimentally validated much better than the horizontal one in terms of HTC and CHF for exploiting boiling heat transfer.
- (2) FeCrAl alloys exhibit better thermal performances over Zircalloys under both steady-state and power transient conditions.
- (3) the post-CHF oxidized samples render higher CHF than as-received counterparts for ATF and traditional claddings.
- (4) under the Fuchs-RIA accident scenarios, MHF is a better indicator of thermal safety margin than CHF that is usually featured with surface temperature overshooting.
- (5) the power-transient MHF values of claddings are higher than their steady-state counterparts and faster power transients with shorter pulse width result in higher pool boiling MHF.
- (6) the progressive decreasing of power transient time scales gradually closes the difference gap of pool boiling MHF between different cladding materials.

Besides FeCrAl superiority of thermal safety margins over Zircalloys, FeCrAl alloys also show better corrosion resistance to surface temperature overshooting and illustrate better structural integrity under the Fuchs-RIA pulse impacts.

4.2. Results implications to ATF-LWR cores

The long time exposure of ATF claddings to extreme environments of LWRs could diminish the CHF difference gaps between different ATF candidates. It speaks to the possibility that the long-term evaluations of ATF claddings should be assessed in material-mechanics, corrosion behaviors and neutronics. The use of present steady-state CHF database could be conservatively safe in the thermal safety margin evaluations of ATF-loaded LWRs under the Fuchs-RIA progression impacts. Under the fast power transients, the heat transfer from solid to liquid would be dominated by transient heat conduction instead of heat convection. This could much more simplify the analytical modeling of the fuel-clad thermal-hydraulics dynamics.

CRedit authorship contribution statement

Mingfu He: Methodologies, Experimental investigation, Data curation, Formal analysis, Visualization, Software, Writing – original draft. **Minghui Chen:** Supervision, Conceptualization, Resources, Funding acquisition, Project administration, Writing – review & editing.

Declaration of competing interest

Authors declare there is no conflict of competing interests.

Data availability

Data will be made available on request

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